# Outline

1. Progress Update
2. Introduction
3. Problem Setting
4. Related Work
5. Expected Methods
6. Experiments Setting
7. Experiments Results
8. Next Steps

# Dataset Overview

1. Dataset:
    - Kaggle Playground S5E8 (Bank Marketing Subscription)
    - https://www.kaggle.com/competitions/playground-series-s5e8/overview
2. Source:
    - Synthetic version of UCI Bank Marketing (Moro et al., 2014)
3. Size: 750k train, 250k test
4. Features:
    - Categorical: job, marital, education, housing, loan, contact type, etc.
    - Numerical: age, balance, duration, previous contacts, etc.
5. Target: subscribed (binary Yes/No)

# Key Observations

- No missing values, all columns clean and consistent.
- Highly imbalanced target: positive (≈11%) vs. negative (≈89%).
- Strongest signal: duration positively correlated with subscription.
- Most categorical variables weakly correlated, implying non-linear patterns.

# Preprocessing Steps

- Convert categorical variables to category
  - For efficient memory and encoding
  - Solve challenge: Model Generalization

```
# 1) cast to category
for df in (train, test, orig):
    for c in CATS:
        df[c] = df[c].astype('category')
```

# Preprocessing Steps

- Normalize / log-transform skewed numeric variables
  - balance_log, duration_sin/cos, pdays_sin
  - Stabilize distribution
  - Solve challenge: Overfitting Risk and Model Generalization

```python
def add_num_transforms(df):
    # log with sign for balance
    df['_balance_log'] = (np.sign(df['balance']) * np.log1p(np.abs(df['balance']))).astype('float32')
    # cyclical for duration (例：以 540s 為週期，可依資料調整)
    df['_duration_sin'] = np.sin(2*np.pi * df['duration'] / 540).astype('float32')
    df['_duration_cos'] = np.cos(2*np.pi * df['duration'] / 540).astype('float32')
    # cyclical for pdays (例：以 365 天對應週期)
    df['_pdays_sin'] = np.sin(2*np.pi * np.clip(df['pdays'], -1, 365) / 365).astype('float32')
    return df
```

# Preprocessing Steps

- Create interaction features (numeric bigrams)
  - Capture non-linear effects between variables.
  - Solve challenge: Weakly Correlated

```
from itertools import combinations

NUMS_ENH = NUMS + ['_balance_log','_duration_sin','_duration_cos','_pdays_sin']

def add_numeric_bigrams(df, cols, cap=32):
    pairs = list(combinations(cols, 2))[:cap]
    for a, b in pairs:
        df[f'BIG_{a}__{b}'] = (df[a].astype('float32') * df[b].astype('float32'))
    return df
```

# Preprocessing Steps

- Apply Target Encoding (TE) and Count Encoding (CE)
  - Uing both competition and original UCI dataset
  - Enriched categorical representations.
  - Solve challenge: Class Imbalance

```
# Count Encoding from UCI
for c in CATS:
    vc = orig[c].value_counts()
    train[f'CE_ORIG_{c}'] = train[c].map(vc).fillna(0).astype('float32')
    test[f'CE_ORIG_{c}']  = test[c].map(vc).fillna(0).astype('float32')
```

# Preprocessing Steps

- Apply Target Encoding (TE) and Count Encoding (CE)
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  - Solve challenge: Class Imbalance

```
from cuml.preprocessing import TargetEncoder

def apply_te_in_fold(X_tr, y_tr, X_va, X_te, cats, seed=42):
    # 只在當前 fold 的訓練資料上 fit；val/test 僅 transform（避免洩漏）
    for c in cats:
        enc = TargetEncoder(n_folds=10, smooth=0, seed=seed,
split_method='random', stat='mean')
        X_tr[c] = enc.fit_transform(X_tr[c], y_tr)
        X_va[c] = enc.transform(X_va[c])
        X_te[c] = enc.transform(X_te[c])
    return X_tr, X_va, X_te
```

# Preprocessing Steps

- Ensure no data leakage
  - Transformations using y (e.g., TE, SMOTE) are fitted on training folds only
  - External statistics (e.g., CE from orig) are computed offline without target labels.
  - Solve challenge: Overfitting & Generalization

# Modeling Strategy

- Explore both tree-based and deep tabular architectures
  - Ensure diversity and robustness.
- Use Stratified 5-fold Cross-Validation for consistent evaluation across all models.
- Optimize hyperparameters with early stopping.
  - learning rate, embedding dim, dropout, epochs

# Models for Feature-Weight Analysis

| Model Type | Primary Method | Notes |
|---|---|---|
| **Boosting / Tree Models** (XGBoost, LightGBM, CatBoost, Random Forest) | **Gain-based importance** for initial ranking,followed by **Permutation Importance** for verification | Provides strong baseline; balances predictive power and interpretability. |
| **Deep Tabular Models** (TabM, FTTransformer, TabR, GRN, CNN, RealMLP, DeepTables) | **Permutation Importance (AUC drop)** | Unified evaluation metric across all architectures for fair comparison. |
| **Ensemble Layer (Blender)** (CatBoost meta-model) | **CatBoost feature importance on base-model scores** | Quantifies each base model's contribution to the final ensemble. |

# Implementation Plan

- Begin with TabM (TabMini architecture) as the first case study.
- Apply Permutation Importance to analyze the influence of each feature group (TE, CE, cyclical, numerical).
- Extend the same process to other models (XGB, LGBM, CatBoost, etc.).

# Evaluation Metrics

| Metric | Description | Purpose |
|---|---|---|
| **Accuracy** | Overall percentage of correct predictions. | General baseline indicator. |
| **Precision** | Proportion of predicted positives that are true positives. | Measures false-positive control. |
| **Recall (Sensitivity)** | Proportion of true positives that are correctly detected. | Important for minority class detection. |
| **F1-score** | Harmonic mean of Precision and Recall. | Balances precision–recall trade-off for imbalanced data. |
| **AUROC** | Area under the ROC curve. | Primary metric for Kaggle; robust to class imbalance. |

# Outline

1. Progress Update
2. Introduction
3. Problem Setting
4. Related Work
5. Expected Methods
6. Experiments Setting
   - 6.1. Dataset
   - 6.2. Evaluation Strategy
7. Experiments Results
8. Next Steps

# Data Description and Characteristics

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| # | △ | Team | Members | Score | Entries | Last | Solution |
|---|---|---|---|---|---|---|---|
| 1 | — | Optimistix | | 0.97801 | 132 | 2mo | |
| 2 | ▲ 3 | Mahog | | 0.97794 | 60 | 2mo | |
| 3 | ▲ 1 | Succeeded. Resource recovered | | 0.97790 | 145 | 2mo | |
| 4 | ▼ 2 | Masaya Kawamata | | 0.97790 | 82 | 2mo | |
| 5 | ▼ 2 | Ravi Ramakrishnan | | 0.97784 | 28 | 2mo | |
| 6 | ▲ 9 | Kyr1ll | | 0.97775 | 31 | 2mo | |
| 7 | ▼ 1 | wind1234it | | 0.97769 | 106 | 2mo | |
| 8 | ▲ 3 | DanteTheAbstract | | 0.97768 | 31 | 2mo | |
| 9 | ▲ 1 | Yeshua T | | 0.97768 | 25 | 2mo | |
| 10 | ▼ 1 | Thiago Lima Santos | | 0.97767 | 45 | 2mo | |

# Dataset Checklist

1. Domain:
   - ✓ **Finance & Economics**
2. Source:
   - ✓ **Public** Kaggle competition (leaderboard & baselines available)
3. Size:
   - ✓ **1M** samples (≫ 5k minimum)
4. Features:
   - ✓ Clear semantics (age, job, contact, etc.) → interpretable
5. Task:
   - ✓ **Binary classification** (Yes/No subscription)

# Basic Exploratory Data Analysis (EDA)

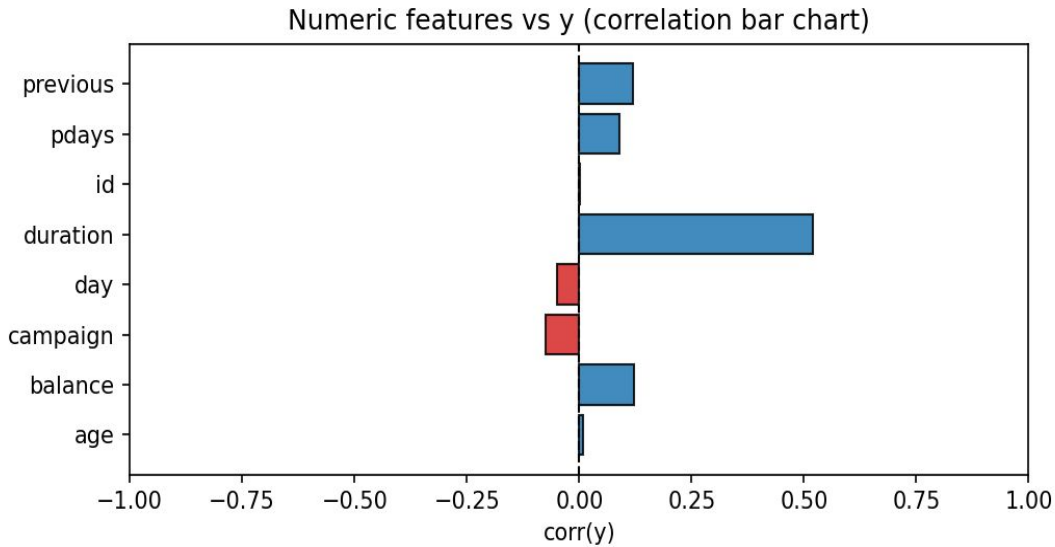
| Variable | Description | Type / Values |
|---|---|---|
| **age** | Client’s age | Numeric |
| **balance** | Average yearly balance (EUR) | Numeric |
| **day** | Last contact day of month | Numeric (1–31) |
| **duration** | Last contact duration (seconds) | Numeric |
| **campaign** | Number of contacts in this campaign | Numeric |
| **pdays** | Days since last contact (-1 = never contacted) | Numeric |
| **previous** | Number of contacts before this campaign | Numeric |
| **y** | Subscribed to term deposit? (target) | Binary: yes, no |

# Basic Exploratory Data Analysis (EDA)

| Variable | Description | Type / Values |
|---|---|---|
| **job** | Type of job | Categorical: admin., blue-collar, entrepreneur, etc. |
| **marital** | Marital status | Categorical: married, single, divorced |
| **education** | Education level | Categorical: primary, secondary, tertiary, unknown |
| **default** | Has credit in default? | Categorical: yes, no |
| **housing** | Has a housing loan? | Categorical: yes, no |
| **loan** | Has a personal loan? | Categorical: yes, no |
| **contact** | Contact communication type | Categorical: unknown, telephone, cellular |
| **month** | Last contact month | Categorical: jan, feb, …, dec |
| **poutcome** | Outcome of previous campaign | Categorical: unknown, other, failure, success |

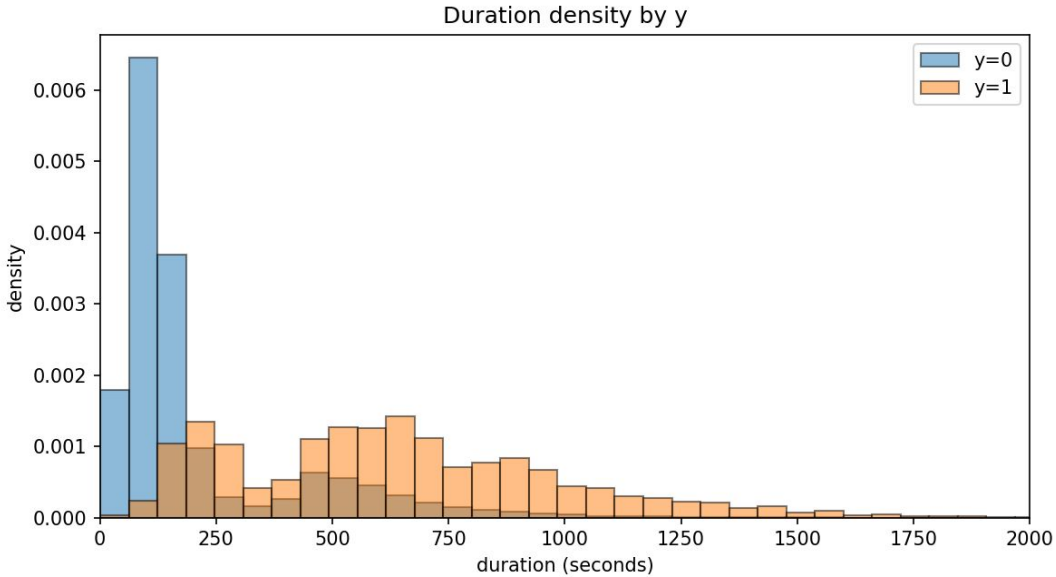
# Basic Exploratory Data Analysis (EDA)

1. missing value rate: 0%
2. numeric vs y correlation analysis
   - duration is strongest signal



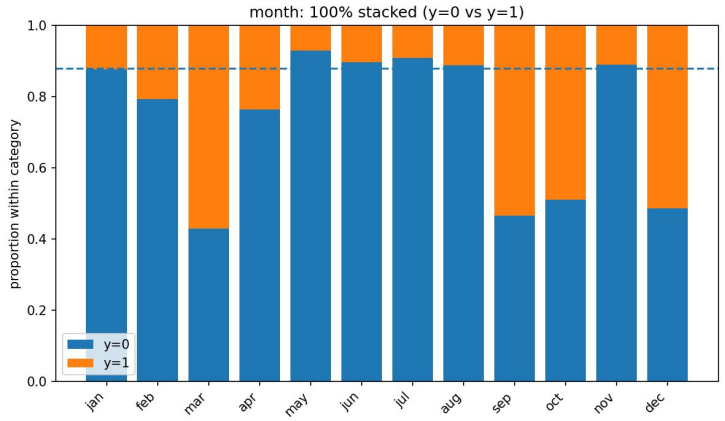
# Basic Exploratory Data Analysis (EDA)

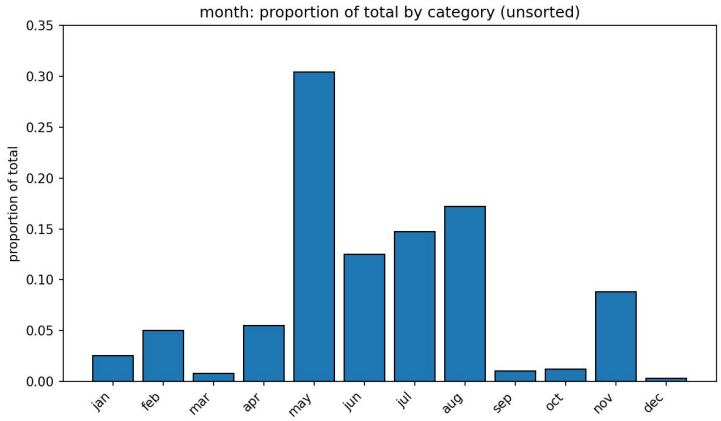
- numeric vs y correlation analysis (duration)
  - Interested clients talk longer



# Basic Exploratory Data Analysis (EDA)

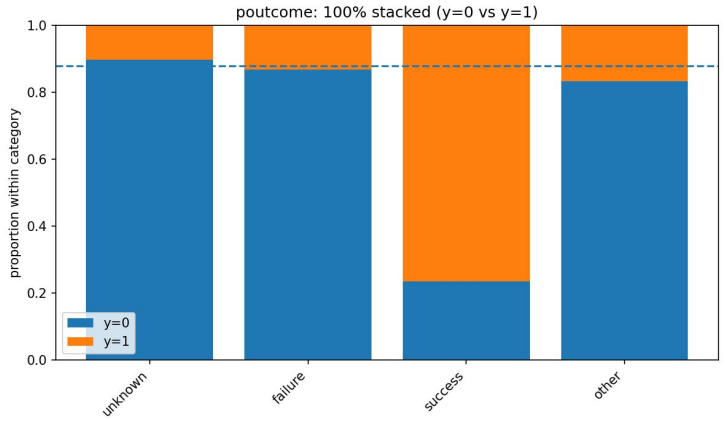
- Analyze non-numeric data (month)
  - In Mar, Sep, Oct, Dec, the proportion of true is high
  - However, their overall share of data is relatively low

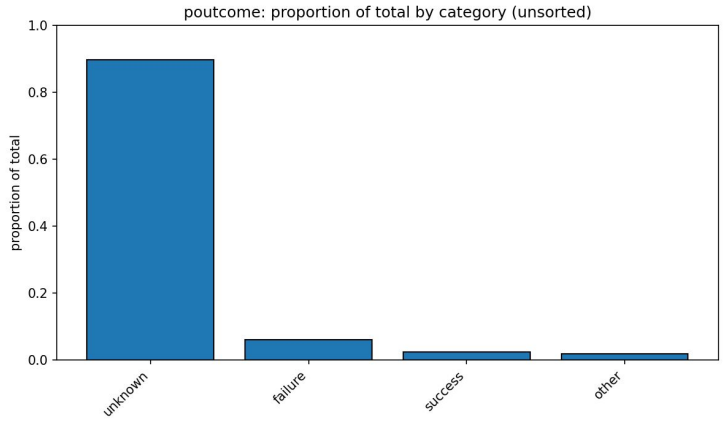


# Basic Exploratory Data Analysis (EDA)

- Analyze non-numeric data (poutcome)
  - If the client had a previous successful outcome,
  - The probability of success is much higher
  - However, this category has very few samples



# Basic Exploratory Data Analysis (EDA)

- Summary:
  - No missing data
  - Most categories show no clear correlation with y
  - Correlated categories cover only a small portion of data

# Basic Exploratory Data Analysis (EDA)

- Dataset clean and usable → feasible for modeling
- Low correlation → signals are non-linear patterns
- Plan:
    a. Create interaction & composite features
    b. Use ensemble models for hidden relationships
    c. Apply SHAP for interpretability

# Synthetic vs. Original Dataset

- Purpose:
  - To expand the training dataset and improve model generalization.
- Difference:
  - Generated by a deep model, so feature distributions are similar but not identical.
  - May alter rare patterns or correlations among variables.
- Potential Distortion:
  - Synthetic process may introduce bias or loss of realism, especially in edge cases.

# Evaluation Strategy

- Use Stratified 5-Fold CV
- Evaluate each fold independently and compute OOF (Out-of-Fold) scores
- Compare cross-model results (TabM, XGB, LGBM, etc.) using the same metrics.
- Main evaluation metrics: AUROC (primary) and F1-score (secondary).
  - Solve Challenge: Class Imbalance

# Outline

1. Progress Update
2. Introduction
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5. Expected Methods
6. Experiments Setting
7. Experiments Results (running)
8. Next Steps

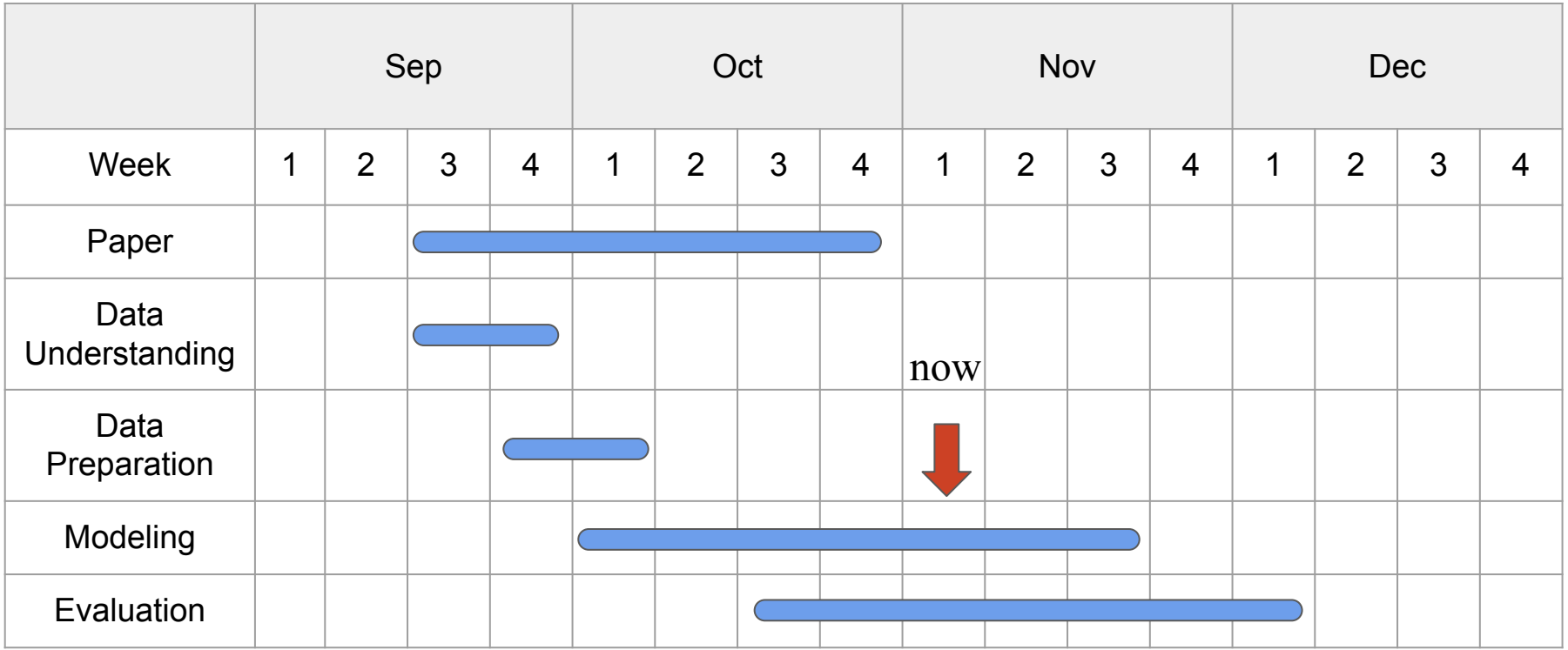
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# Next Steps - Schedule



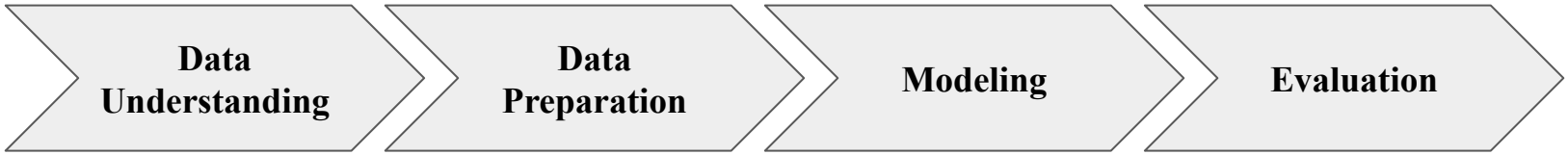
# Environment

1. OS
    - Linux / Windows
2. Programming Language
    - Python
3. Tools
    - VSCode
4. Libraries
    - pytorch, pandas, numpy, dgl

# Analysis Workflow

- Explore dataset
- Check quality
- Analyze imbalance
- Distributions

- Train baseline & advanced ML models (tree-based, ensemble, DL)
- Tune hyperparameters




- Clean & preprocess data (encoding, scaling)
- Handle class imbalance – apply SMOTE on training set only
- Feature engineering & selection

- Use metrics (Accuracy, AUROC, F1, PR-AUC)
- Apply stratified k-fold CV & hold-out test